EECE 5639: Computer Vision
Fall 2026, Homework 4

Assigned: Sun, Nov 01, 2026 **Points: 100** **Due: Mon, Nov 09, 2026**

1. **[20 points]** The Harris Corner Detector uses the second-moment matrix

$$M = \sum_{x,y} w(x,y) \begin{bmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{bmatrix} = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$$

to characterize the local intensity variations in an image. The Harris response is defined as $R = \det(M) - k[\text{trace}(M)]^2$, where $k = 0.04$. Consider the following second moment matrices corresponding to five different local regions in an image:

$$M_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad M_2 = \begin{bmatrix} 100 & 0 \\ 0 & 1 \end{bmatrix}, \qquad M_3 = \begin{bmatrix} 1 & 0 \\ 0 & 100 \end{bmatrix}$$

$$M_4 = \begin{bmatrix} 100 & 0 \\ 0 & 50 \end{bmatrix}, \qquad M_5 = \begin{bmatrix} 87.5 & 12.5\sqrt{3} \\ 12.5\sqrt{3} & 62.5 \end{bmatrix}.$$

(a) **[5 points]** Calculate the Harris response R for each of the five matrices using $k = 0.04$. Show your calculations.

(b) **[5 points]** Based on the Harris responses from part (a) and the eigenvalues of the second moment matrices, classify each region as one of the following:

- Flat region
- Edge
- Corner

(c) **[3+2 = 5 points]** Based on the eigenvalues and eigenvectors of M_2 and M_3, compare the magnitudes and directions of the local intensity variations in these two regions. Explain why the two regions produce identical Harris responses despite their different orientations.

(d) **[3+2 = 5 points]** Based on the eigenvalues and eigenvectors of M_4 and M_5, compare the magnitudes and directions of the local intensity variations in these two regions. Explain what your observations reveal about the rotation invariance of the Harris Corner Detector.

2. **[30 points]** Consider the following 5×5 grayscale image, which contains a triangular structure. The intensity is 200 inside the structure and 0 elsewhere.

$$I_1 = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 200 & 0 & 0 \\ 0 & 200 & 200 & 200 & 0 \\ 200 & 200 & 200 & 200 & 200 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

The tip of the triangular structure is located at $P = (2, 3)$, where coordinates are specified as $(\text{row}, \text{column})$, using one-based indexing.

The second-moment matrix is

$$M = \sum_{x,y} w(x, y) \begin{bmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{bmatrix}.$$

We will use the 3×3 Gaussian window:

$$w = \frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}.$$

Approximate the image gradients using central differences:

$$I_x(r, c) = \frac{I(r, c+1) - I(r, c-1)}{2},$$

$$I_y(r, c) = \frac{I(r+1, c) - I(r-1, c)}{2}.$$

For pixels on the image boundary, use zero-padding when computing the gradients.

The Harris response is

$$R = \det(M) - k[\text{trace}(M)]^2, \qquad k = 0.04.$$

(a) **[3 points]** Calculate the image gradients I_x and I_y at every pixel in the 3×3 neighborhood centered at $P = (2, 3)$.

(b) **[3 points]** Using these gradients and the Gaussian weighting matrix, calculate the second-moment matrix M_1 and the Harris response R_1.

(c) **[3 points]** Based on the sign and magnitude of R_1, classify the pixel as corner-like, edge-like, or flat.

(d) **[7 points]** Consider a second image:

$$I_2 = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 100 & 0 & 0 \\ 0 & 100 & 100 & 100 & 0 \\ 100 & 100 & 100 & 100 & 100 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

(i) **[5 points]** Without repeating all the gradient calculations, determine the second-moment matrix M_2 and the Harris response R_2 at $P = (2, 3)$.

(ii) **[2 points]** Compare R_2 with R_1. Explain the significance of any changes in the magnitude and sign of the Harris response.

(e) **[8 points]** Consider a third image I_3, obtained by enlarging I_1 by a factor of five in both spatial directions using nearest-neighbor interpolation. Each pixel in I_1 is replaced by a 5×5 block of pixels having the same intensity, producing the following 25×25 image. For compactness, the image is displayed in block form, where each entry represents a 5×5 block of identical pixel intensities:

$$I_3 = \begin{bmatrix} \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{B} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{B} & \mathbf{B} & \mathbf{B} & \mathbf{0} \\ \mathbf{B} & \mathbf{B} & \mathbf{B} & \mathbf{B} & \mathbf{B} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix},$$

where

$$\mathbf{0} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \qquad \mathbf{B} = \begin{bmatrix} 200 & 200 & 200 & 200 & 200 \\ 200 & 200 & 200 & 200 & 200 \\ 200 & 200 & 200 & 200 & 200 \\ 200 & 200 & 200 & 200 & 200 \\ 200 & 200 & 200 & 200 & 200 \end{bmatrix}.$$

The original triangular tip at $P = (2, 3)$ corresponds to the block occupying rows 6–10 and columns 11–15. Consider the center of this block: $P' = (8, 13)$.

(i) **[5 points]** Calculate the second-moment matrix M_3 at P', using the same central difference approximations and the same 3×3 Gaussian weighting matrix:.

(ii) **[1 point]** Calculate the Harris response R_3.

(iii) **[2 points]** Compare R_3 with R_1. How would you classify the pixel P' based on its Harris response?

(f) **[2 points]** Explain why the Harris response at the center of the enlarged tip differs from the response at the original tip.

(g) **[2 points]** Does the result in part (e) imply that the triangular corner has disappeared from the enlarged image? Explain.

(h) **[2 points]** Explain why examining an image at multiple scales can help address the limitations of using a fixed-size neighborhood for corner detection.

3. **[20 points]** Consider the two triangular images I_1 and I_3 from Question 2. We will now investigate how the Difference-of-Gaussians (DoG) detector used in SIFT identifies image features at different spatial scales.

The Gaussian scale-space representation is $L(x, y, \sigma) = G(x, y, \sigma) * I(x, y)$, where

$$G(x, y, \sigma) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{x^2 + y^2}{2\sigma^2}\right).$$

The Difference-of-Gaussians (DoG) is $D(x, y, \sigma) = L(x, y, k\sigma) - L(x, y, \sigma)$, where $k > 1$.

For this problem, use Python or MATLAB to perform the calculations. Use the same Gaussian filtering and boundary-handling conventions for both images. Use a sufficiently fine sequence of scales to identify local extrema. For example, you may use $\sigma_j = 0.5(2^{1/8})^j$, with enough values of j to cover scales up to $\sigma \approx 6$. You may use built-in Gaussian filtering functions, but you must construct the DoG images and implement the scale-space extremum search yourself.

(a) **[5 points]** Construct the Gaussian scale-space representations of I_1 and I_3. Use reflective boundary handling when applying Gaussian filtering to both images.. Display representative Gaussian-smoothed images at several values of σ.

(b) **[5 points]** Compute the DoG images using consecutive Gaussian scales:

$$D_j(x, y) = L(x, y, \sigma_{j+1}) - L(x, y, \sigma_j).$$

Display representative DoG images for both I_1 and I_3.

(c) **[8 points]** A candidate DoG keypoint must be a local maximum or minimum relative to its 26 neighbors: eight neighbors at the same scale, nine at the preceding scale, and nine at the following scale.

(i) Implement the 26-neighbor comparison to identify local extrema in the DoG scale space. Include your code in the ZIP file submitted with this homework.

(ii) Report the locations, scales, and DoG values of the detected extrema in both images.

(iii) Identify a pair of extrema, if available, that correspond to the same triangular image structure. Report their characteristic scales, σ_1 and σ_3.

(iv) Calculate the ratio $\frac{\sigma_3}{\sigma_1}$. Compare this ratio with the spatial enlargement factor of five.

(v) If no suitable corresponding pair is detected, explain your observations using the scale-space responses.

(d) **[2 points]** Explain how the scale-selection mechanism differs from the fixed-size Harris detector used in Question 2.

4. **[30 points]** Consider the two images shown in Figure 1. They are provided as attachments `001.jpg` and `002.jpg` with this homework. Both of them depict the same building, but from different perspectives. Let I_1 denote the original image (left) and I_2 denote the transformed image (right). We want to estimate a homography that maps points in the original image I_1 to their corresponding locations in the transformed image I_2.

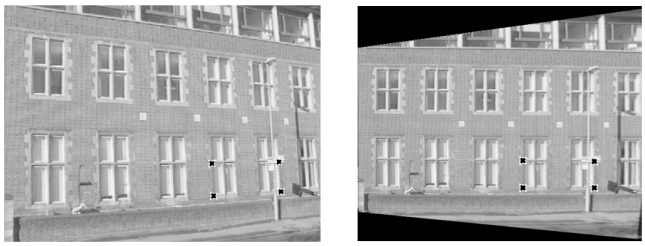


Figure 1: Perspective Correction Using Homography

Recall that a homography relates corresponding homogeneous pixel coordinates through

$$\begin{bmatrix} u_2 \\ v_2 \\ 1 \end{bmatrix} \sim H \begin{bmatrix} u_1 \\ v_1 \\ 1 \end{bmatrix},$$

where $H = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix}$. The symbol $\sim$ denotes equality up to a nonzero scale factor. We are interested to find the homography matrix H such that $\sum_{i,j} h_{ij}^2 = 1$. You may use Python or MATLAB to complete this problem.

(a) **[5 points]** Manually select 20 accurate candidate point correspondences between I_1 and I_2. For each correspondence, record

$$(u_{1i}, v_{1i}) \longleftrightarrow (u_{2i}, v_{2i}).$$

for $i = 1, 2, ..., 20$. Display the correspondences on the two images using matching markers.

(b) **[5 points]** Using the candidate correspondences, estimate the homography H that maps points from I_1 to I_2. That is, compute the data matrix A satisfying $Ah = 0$, then determine the vector h which minimizes the algebraic distance $||Ah-0||_2^2$ subject to $||h||^2 = 1$. You may use a numerical linear algebra routine to solve the resulting system. Report the estimated homography matrix H.

(c) **[14 points]** In parts (a) and (b), you identified point correspondences between I_1 and I_2 and estimated a homography using the "Direct Linear Transform" (DLT) method. We will now investigate how incorrect correspondences (outliers) affect homography estimation.

(i) **[2 points]** Randomly select 25% of the point correspondences from part (a). Replace their correct target matches with randomly selected pixel locations in I_2. Keep the remaining correspondences unchanged. Display the resulting correspondences and clearly identify the deliberately introduced outliers.

(ii) **[2 points]** Using all the correspondences, including the introduced outliers, estimate a new homography H_{outlier} using DLT.

(iii) **[7 points]** Apply RANSAC to the correspondences containing outliers. Use the adaptive version of RANSAC we learnt for line fitting, but this time the sample size is $s = 4$. In each iteration:

A. Randomly select four correspondences.
B. Estimate a homography using DLT.
C. Transform the source-image points using the estimated homography.
D. Calculate the reprojection error

$$e_i = \sqrt{(u_{2i} - \hat{u}_{2i})^2 + (v_{2i} - \hat{v}_{2i})^2},$$

where $(\hat{u}_{2i}, \hat{v}_{2i})$ is the predicted location of the ith point in I_2.
E. Classify a correspondence as an inlier if $e_i < \tau$, where $\tau = 3$ pixels.

In the end, select the homography with the largest inlier consensus set. Re-estimate the homography using all identified inliers. Denote the resulting homography by H_{RANSAC}. Include your RANSAC implementation in the ZIP file submitted with the homework..

(iv) **[3 points]** Report the following:

- The number of iterations
- The final homography matrix H_{RANSAC}.
- The number of inliers and outliers.

In addition, display the correspondences classified as inliers and outliers using different colors on the images I_1 and I_2.

(d) **[6 points]** We have now obtained three homography estimates:

- H_{DLT}: Estimated using the original, manually identified correspondences from part (a).
- H_{outlier}: Estimated using all correspondences, including the deliberately introduced outliers.
- H_{RANSAC}: Estimated using RANSAC followed by DLT on the identified inliers.

We will compare how accurately these homographies transform I_1 into the coordinate system of I_2.

(i) **[2 points]** Warp I_1 using each of the three homographies. Denote the resulting images by $\hat{I}_{2,\text{DLT}}$, $\hat{I}_{2,\text{outlier}}$, $\hat{I}_{2,\text{RANSAC}}$ respectively. Display the three warped images alongside the ground-truth target image I_2.

(ii) **[2 points]** Evaluate the similarity between each warped image and the ground-truth image using Mean Squared Error (MSE):

$$\text{MSE} = \frac{1}{|\Omega|} \sum_{(u,v)\in\Omega} \left[I_2(u,v) - \hat{I}_2(u,v) \right]^2,$$

where Ω denotes the set of valid pixels covered by both images. For this problem, use the fixed rectangular evaluation region $\Omega = \{(u,v) : 100 \le u \le 350,\ 100 \le v \le 250\}$. Use this same region for all three homography estimates. If a warped pixel has no valid source-image correspondence, assign it an intensity of zero.

Report your results in the following table:

Homography Estimation Method	**MSE**
DLT (original correspondences)	
DLT (with outliers)	
RANSAC + DLT	

(iii) **[2 points]** Using the MSE values from part (d)(ii), explain how the presence of incorrect correspondences affects the homography estimation, and how RANSAC helps address this problem.